

UPS Frontend – Technical Documentation

Version: 0.1.0
Last Updated: February 10, 2026

Table of Contents

- [1. Project Overview](#)
- [2. Deployment Environment](#)
- [3. Project Structure](#)
- [4. Architecture & Design Patterns](#)
- [5. Core Components Documentation](#)
- [6. State Management](#)
- [7. Routing](#)
- [8. Styling System](#)
- [9. API Integration](#)
- [10. Internationalization \(i18n\)](#)
- [11. Authentication & Authorization](#)
- [12. Build & Deployment](#)
- [13. Performance Optimizations](#)
- [14. Accessibility \(a11y\)](#)
- [15. Security Considerations](#)
- [16. Troubleshooting Guide](#)

1. Project Overview

Application Name & Version

Admin Dashboard Design v0.1.0

A comprehensive water infrastructure monitoring and management system.

Description

A React-based dashboard providing real-time monitoring and management for water pumping stations and water level monitoring sites. Administrators and operators can track readings, configure alarms, manage sites, and monitor system health across multiple directorates.

Key Features

- Multi-role access control (Admin / Operator)
- Real-time dashboard monitoring
- Site CRUD management
- Flexible alarm configuration (Email, SMS, WhatsApp)
- Historical readings and logs
- Formula-based flow calculations
- Arabic & English localization with RTL support
- Responsive, mobile-first design

User Benefits

- Centralized infrastructure monitoring
- Proactive alarm handling
- Data-driven operational insights
- Secure role-based access
- Reduced training overhead

Technology Stack Summary

Core Framework

- React 18.3.1
- TypeScript (Vite)
- React Router DOM 7.9.5

Build Tool

- Vite 6.4.1 (SWC)

UI Framework

- Tailwind CSS 4.1.17
- Radix UI
- shadcn/ui
- Lucide React Icons

State & Data

- React Context API
- Custom hooks
- Axios 1.13.2

Forms

- React Hook Form 7.55.0
- Formik 2.4.9
- Yup 1.7.1

Charts

- Recharts 2.15.2

Utilities

- i18next 25.7.3
- js-cookie 3.0.5
- jwt-decode 4.0.0
- react-toastify 11.0.5
- xlsx 0.18.5

2. Deployment Environment

Browser Requirements

- Chrome ≥ 90
- Firefox ≥ 88
- Safari ≥ 14
- Edge ≥ 90

Deployment Platform

- **Platform:** Vercel
- **Type:** SPA deployment with history fallback

Environment Configuration

```
const API_BASE_URL = 'https://tele-dairot:5050/api';
```

Build Configuration

- Vite + SWC
- Path alias: @ → ./src
- Tailwind + Autoprefixer

3. Project Structure

Directory Tree (src)

```
src/
├── features/
├── components/
├── shared/
├── i18n/
├── lib/
├── styles/
├── App.tsx
├── main.tsx
└── index.css
```

Feature-Based Module Pattern

```
feature-name/
├── components/
├── hooks/
├── types/
├── utils/
├── services/
└── index.ts
```

Benefits

- High cohesion
- Clear ownership
- Scalable structure
- Easier onboarding

4. Architecture & Design Patterns

System Architecture

```
graph TB
  Browser --> main.tsx --> App.tsx --> AuthProvider --> Router --> DashboardLayout --> Features
  Features --> apiService --> Backend
```

Patterns Used

- Feature-based architecture
- Atomic design principles
- Centralized API service
- Context + hooks state model

5. Core Components Documentation

5.1 LoginPage

Location: src/features/auth/components/LoginPage.tsx

Purpose: User authentication.



English ▾

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

Email

admin@gmail.com

Password

.....

Login

Secure and encrypted connection

Features

- Credential login
- Validation
- Remember me
- i18n support

5.2 DashboardLayout

Location: src/components/DashboardLayout.tsx

Purpose: Application shell and navigation.

```
interface DashboardLayoutProps {
  currentUser: User;
  onLogout: () => void;
  refreshCurrentUser: () => void;
}
```

Features

- Responsive sidebar
- Role-based menus
- RTL support
- <Outlet /> routing

5.3 DashboardHome

Admin only

Main Dashboard

Overview of the National Irrigation Network

Total Sites

45

All sites connected

Active Alarms

0

3 Critical • 9 Warnings

Total Directorates

5

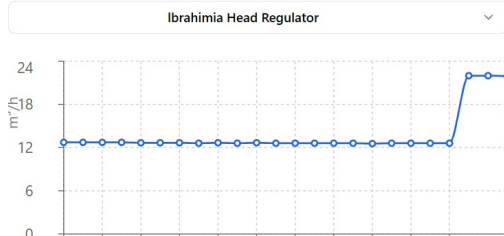
All Regions Active

Total Users

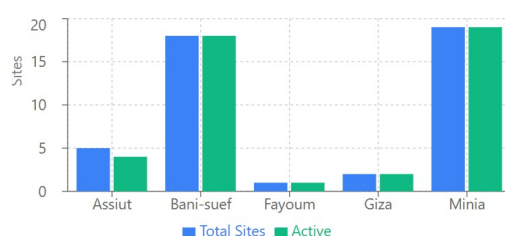
5

System Users

Flow in Last 24 Hours



Sites by Directorate



Features

- KPI cards
- Charts
- Alarm summaries
- Reading logs

5.4 SitesManagement

CRUD + filtering

Sites Management

Manage water level monitoring sites and pump stations

Search for a site...



All Types



All Directorates



All Canals



Sites (45)

[+ Add New Site](#)

Site Name	Site Type	SIM ID	Data Logger Type	Canal	Directorate	Location	Actions	
 El-Kosia	Water Level		CR350	Ibrahimia	Assiut			
 Irad El Maharak	Water Level		CR350	Ibrahimia	Assiut			
 Ibrahimia Head Regulator	Water Level		CR350	Ibrahimia	Assiut			
 test	Water Level	10.100.10.0	CR310	Ibrahimia	Assiut			

```
useSitesData()
```

Features

- Search & filters
- Dialog forms
- Pagination
- Role-based actions

5.5 Readings Management

- Water level readings
- Pump readings
- History & export

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Readings Management

Manage pump station and water level readings

Ibrahimia Head Regulator

December 1st, 2025February 1st, 2026X

Water LevelPump Station

Water Level Readings (10)

ExportAdd Manual Reading

Site	Date and Time	USWL (meter)	DSWL1 (meter)	Battery (V)	Calculated Flow	Actions
Ibrahimia Head Regulator	26/12/2025, 16:00:00	49.9700	48.2300	13.75	21.92 m³/h	
Ibrahimia Head Regulator	26/12/2025, 15:00:00	49.9600	48.2400	13.91	21.99 m³/h	
Ibrahimia Head Regulator	26/12/2025, 14:00:00	49.9700	48.2400	13.93	21.99 m³/h	
Ibrahimia Head Regulator	26/12/2025, 13:00:00	28.0900	28.6400	13.97	12.61 m³/h	

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Readings Management

Manage pump station and water level readings

New Kamdeer PS

October 1st, 2025February 1st, 2026X

Water LevelPump Station

Pump Station (6)

ExportAdd Manual Reading

Site	Date and Time	Total Uptime	Total Flow	Actions
New Kamdeer PS	11/01/2026, 03:00:00	14.0 hour	17.0 m³/h	
New Kamdeer PS	26/10/2025, 09:00:00	237.0 hour	601.8 m³/h	
New Kamdeer PS	25/10/2025, 14:00:00	239.0 hour	602.9 m³/h	
New Kamdeer PS	24/10/2025, 10:00:00	234.0 hour	595.8 m³/h	

5.6 ReadingLogs

- Chronological logs
- Filters
- Excel export

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Reading Logs

View all reading logs and actions

Search by site name, action, reading

mm/dd/yyyy

Search

Records per page10

Reading Logs (0)

ID	Reading Type	Action Type	Site Name	Action Date	Timestamp	Created By
14	WaterLevel	Deleted	test	09/02/2026, 14:01:29	2026-02-09T02:00:00	admin@gmail.com
13	WaterLevel	Created	test	09/02/2026, 14:01:20	2026-02-09T02:00:00	admin@gmail.com
12	WaterLevel	Created	New Kamdeer PS	09/02/2026, 13:57:19	2026-02-09T01:00:00	admin@gmail.com
11	WaterLevel	Updated	test eqn	04/02/2026, 13:12:51	2026-02-04T13:00:00	admin@gmail.com
10	WaterLevel	Updated	test eqn	04/02/2026, 13:10:45	2026-02-04T13:00:00	admin@gmail.com
9	WaterLevel	Updated	test eqn	04/02/2026, 13:03:48	2026-02-04T13:00:00	admin@gmail.com
8	WaterLevel	Created	test eqn	04/02/2026, 13:01:23	2026-02-04T13:00:00	admin@gmail.com

5.7 AlarmConfiguration

- Threshold alarms
- Communication loss
- Sensor & pump status
- Multi-channel notifications

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

Alarm Configuration

Manage alarm settings

Threshold Alarms

Communication Loss Alarms

Sensor Status

Pump Status (PS)

Individual Pump Status

Threshold Alarms (3)

+ Add Threshold Alarm

Alarm Name	Site	Crisis	Critical	Field	Recipients	Actions
منسوب حرج الامامي	New Hafze Regulator	< 0	<= 42	DSWL1	01066632353	
new alarm	Ibrahimia Head Regulator	>= 2.6	> 2	Total Uptime	user@email.com	
test1	Ibrahimia Head Regulator	< 0.6	< 3	USWL	abdalmalkasam2131@gmail.com	

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

Alarm Configuration

Manage alarm settings

Threshold Alarms

Communication Loss Alarms

Sensor Status

Pump Status (PS)

Individual Pump Status

Communication Loss Alarms (1)

+ Add Communication Loss Alarm

Alarm Name	Site	Hours	Recipients	Actions
فقدان الاتصال	Ibrahimia Head Regulator	1 Hour	01066632353	

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

Alarm Configuration

Manage alarm settings

Threshold Alarms

Communication Loss Alarms

Sensor Status

Pump Status (PS)

Individual Pump Status

Sensor Status (1)

+ Add Sensor Status Alarm

Alarm Name	Site	Field	Threshold Value	Message	Recipients
new alarm	Dahab Regulator	USWL	888	test dahb	noleh73025@codgal.com

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Alarm Configuration

Manage alarm settings

Threshold Alarms

Communication Loss Alarms

Sensor Status

Pump Status (PS)

Individual Pump Status

Configure alarms for individual pump status (1)

+ Add Individual Pump Status Alarm

Alarm Name	Site	Pump number	Recipients	Monitoring Hours	Actions
p1 new kamdeer	New Kamdeer PS	Pump 1	admin@email.com	24 Hours	

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Alarm Configuration

Manage alarm settings

Threshold Alarms

Communication Loss Alarms

Sensor Status

Pump Status (PS)

Individual Pump Status

Configure alarms for pump status (1)

+ Add Pump Status(PS) Alarm

Alarm Name	Site	Duration	Recipients	Actions
test	New Kamdeer PS	12	abdalmalkasama15@gmail.com	

5.8 AlarmEvents

- Live alarm feed
- Acknowledgement
- Prioritization

- Dashboard
- Sites Management
- Readings
- Reading logs
- Alarm Configuration
- Alarm Events
- Alarm Reports
- Flow Calculations
- User Management

Alarm Events

Alarms automatically generated from sensor readings and pump station data

Ibrahimia Head Regulator

From Date

To Date

Clear Filters

Table View

Cards View

ID	Alarm Name	Site	Field	Value	Threshold Value	Severity	Color	Date & Time
#2157	فقدان الاتصال	Ibrahimia Head Regulator	Communication Loss	1078.17	1	Information		02/09/2026 14:10:22
#2156	فقدان الاتصال	Ibrahimia Head Regulator	Communication Loss	1077.67	1	Information		02/09/2026 13:40:22
#2155	فقدان الاتصال	Ibrahimia Head Regulator	Communication Loss	1077.17	1	Information		02/09/2026 13:10:22
#2154	فقدان الاتصال	Ibrahimia Head Regulator	Communication Loss	1076.67	1	Information		02/09/2026 12:40:22

5.9 AlarmReports

- Date range reports
- Statistics
- Export & scheduling

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

Alarm Reports Configuration

Configure and manage scheduled email reports for alarm events

Create New Configuration

Configuration Name	Status	Frequency	Schedule	Recipients	Fields
Dahab Regulator Sunday Weekly Schedule	Active	Weekly	Sunday : 06:00	mazenabaza022@gmail.com	14 fields
test ibrahimia report	Active	Daily	06:00	mazenabazatest@gmail.com	Id
test	Active	Daily	06:00	Abdelmalek@gmail.com	15 fields
El-kosia daily	Active	Daily	06:00	mazenabaza022@gmail.com, mazenabazasbc@gmail.com	3 fields
test2	Active	Daily	06:00	as@1.com	15 fields
test daily	Active	Daily	06:00	aj@ak.com	PumpStationRe
new kamdeer weekly	Active	Daily	06:00	admin@email.com	TriggeredAt, Fi

5.10 FlowCalculations

- Formula validation
- Real-time & historical data

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

Flow Calculations

Configure and manage flow calculation formulas

Select Site

Ibrahimia Head Regulator

Calculation Method

معادلة

Equation Configuration

Equation: $c1 \cdot \text{DSWL}^2 - c2 \cdot \text{DSWL} + c3$

a (Constant)

0.3134

b (Constant)

23.616

c (Constant)

431.91

Save Constants

5.11 UserManagement (Admin)

- User CRUD
- Role assignment
- Activity tracking

Irrigation Monitoring System

Ministry of Water Resources and Irrigation - Egypt

English

Marwa Admin

Administrator

Dashboard

Sites Management

Readings

Reading logs

Alarm Configuration

Alarm Events

Alarm Reports

Flow Calculations

User Management

User Management

Manage user accounts and permissions

+ Add New User

Users (5)

Full Name	Email	Role	Status	Actions
 kjhgjf hgffddd	marwa_emam83@yahoo.com	Administrator	Inactive	
 Mostafa Salah	m.salah@ups.com.eg	Operator	Active	
 Marwa Admin	admin@gmail.com	Administrator	Active	
 Test Admin	TestAdmin@gmail.com	Administrator	Active	
 Maged Test	maged@gmail.com	Operator	Inactive	

6. State Management

Global State – AuthContext

```
interface AuthContextType {
  isAuthenticated: boolean;
  currentUser: User | null;
  handleLogout(): Promise<void>;
  refreshCurrentUser(): void;
}
```

Local State

- Feature hooks
- UI state
- Form state

Persistence

- Cookies: tokens
- SessionStorage: auth flag
- LocalStorage: language

7. Routing

Strategy

- React Router v7
- Nested routes
- Role-based redirects

```
<Route path="/" element={<DashboardLayout />}>
  <Route index element={<DashboardHome />} />
  <Route path="sites" element={<SitesManagement />} />
</Route>
```

8. Styling System

Tailwind CSS v4

- Utility-first
- CVA variants
- CSS variables

RTL Support

- Dynamic dir
- Font switching
- Icon mirroring

9. API Integration

Axios Service Layer

- Auth interceptors
- Token refresh queue
- Centralized error translation

Example

```
apiService.get<ApiResponse<Site[]>>('/v1/Sites');
```

10. Internationalization (i18n)

Setup

- i18next + react-i18next
- Arabic default
- RTL-aware layout

```
t ('sites.title')
```

11. Authentication & Authorization

Flow

- JWT-based auth
- Refresh tokens
- Cookie storage

Roles

Feature	Admin	Operator
Dashboard	☒	☒
Sites	☒	☒
Users	☒	☒

12. Build & Deployment

Build

```
npm run build
```

Platform

- Vercel
- SPA fallback
- Zero downtime

13. Performance Optimizations

Current

- SWC
- Tree shaking
- Memoization

Recommended

- Route-based lazy loading
- React Query
- Table virtualization
- PWA caching

14. Accessibility (a11y)

Current

- Semantic HTML
- ARIA via Radix
- Keyboard support

Enhancements

- Skip links
- Screen reader announcements
- Explicit icon labels

15. Security Considerations

- Token refresh isolation
- Role-based UI enforcement
- Centralized logout on auth failure

16. Troubleshooting Guide

Issue	Resolution
Infinite 401 loop	Check refresh token expiry
RTL layout breaks	Verify <code>_rtl</code> key
Missing data	Confirm <code>API_BASE_URL</code>
Blank page after deploy	Check SPA fallback